Name: _____

Grade / Class: _____

COMPREHENSIVE MODERN ENGLISH STUDIES TEST

(English • English Literature • English Language & Linguistics • Academic English • Technical English • Business English)

SECTION A: ENGLISH (Language Skills & Usage)

1. **Rewrite the following sentence by changing it into passive voice:**
   The committee rejected the proposal after a long discussion.
2. **Identify the type of sentence and justify your answer:**
   Although the weather was unfavorable, the expedition continued.
3. **Replace the underlined word with a more precise synonym:**
   The scientist made an important discovery during the experiment.
4. **Correct the grammatical errors in the following sentence:**
   Neither of the answers were acceptable to the examiner.
5. **Write a paragraph explaining how tone affects meaning in spoken communication.**

SECTION B: ENGLISH LITERATURE

6. **Explain how setting influences the development of characters in a novel or short story you have studied.**
7. **Analyze the role of conflict in shaping the theme of a literary work.**
8. **Differentiate between a tragic hero and an anti-hero using literary reasoning.**
9. **Discuss how symbolism can convey deeper meanings beyond the literal narrative.**
10. **Write a critical response on how literature reflects social change across time.**

SECTION C: ENGLISH LANGUAGE & LINGUISTICS

11. **Define morphology and explain its importance in understanding word formation.**
12. **Distinguish between phonetics and phonology with suitable examples.**
13. **Explain how semantics differs from pragmatics in language interpretation.**
14. **Analyze the function of syntax in maintaining clarity in complex sentences.**
15. **Describe how language variation occurs due to geographical and social factors.**

SECTION D: ACADEMIC ENGLISH

16. **Write an introduction for a research paper on the impact of digital media on education.**
17. **Explain the difference between a literature review and a discussion section in academic writing.**
18. **Identify and explain two common problems related to plagiarism and how they can be avoided.**
19. **Rewrite the following statement in formal academic style:**
    Many people think online learning is better, but it has some problems.
20. **Discuss the importance of coherence and cohesion in academic essays.**

SECTION E: TECHNICAL ENGLISH

21. **Rewrite the following instruction so it is clear and unambiguous:**
    Start the machine after checking everything.
22. **Explain why precision and clarity are essential in technical documentation.**
23. **Convert the following description into a step-by-step procedure:**
    A user logs into a system, uploads a file, and confirms submission.
24. **Identify two differences between technical reports and general articles.**
25. **Write a short explanation of how diagrams support understanding in technical texts.**

SECTION F: BUSINESS ENGLISH

26. **Write a professional email requesting an extension for submitting a project report.**
27. **Explain the importance of tone and politeness in workplace communication.**
28. **Rewrite the following sentence to make it suitable for a formal business proposal:**
    We want to start a new project because it might make more money.
29. **Discuss how clear language can prevent misunderstandings in contracts and agreements.**

30. **Write a brief paragraph explaining the role of persuasive language in marketing communication.**

SECTION G: APPLIED & CRITICAL THINKING

31. **Compare the language style used in academic writing with that used in business reports.**
32. **Explain how audience awareness changes language choice in technical and non-technical texts.**
33. **Analyze how cultural context influences interpretation of English texts globally.**
34. **Describe the role of critical reading in evaluating written arguments.**
35. **Write a concluding paragraph summarizing why mastery of modern English is essential in higher education and professional life.**

Name: _____

Grade / Class: _____

COMPREHENSIVE ASIAN LANGUAGES TEST

(Urdu, Arabic, Persian, Pashto, Sindhi, Punjabi, Balochi, Hindi, Bengali, Tamil, Telugu, Malayalam, Sinhala, Nepali, Chinese (Mandarin), Japanese, Korean, Thai)

SECTION A: URDU (Language, Literature & Grammar)

1. **Identify the correct form of the verb in the following sentence:**
   لڑکا کتاب ___ رہا ہے۔ (پڑھ / پڑھتا / پڑھی)
2. **Translate into English:**
   محبت انسان کی زندگی کا اہم حصہ ہے۔
3. **Write a short paragraph (5–6 lines) describing your favorite Urdu poem and its theme.**
4. **Identify the figure of speech in the following verse:**
   روشنی نے اندھیروں سے مقابلہ کیا۔
5. **Convert the following sentence into passive voice:**
   استاد نے طلباء کو سبق پڑھایا۔

SECTION B: ARABIC

6. **Translate the following into Arabic:**
   Knowledge is the key to success.
7. **Identify the subject and predicate in the sentence:**
   يقرأ الطالب الكتاب في المكتبة.
8. **Write a short paragraph (3–4 sentences) about the importance of Arabic calligraphy.**
9. **Convert the sentence to negative form:**
   ذهب المعلم إلى المدرسة.
10. **Explain the meaning of the Arabic proverb:**
    من جد وجد.

SECTION C: PERSIAN (Farsi)

11. **Translate into English:**
    زندگی کوتاه است، پس قدر آن را بدان.
12. **Identify the tense in the sentence:**
    من دیروز به بازار رفتم.
13. **Write a 5-line paragraph about a famous Persian poet and their contribution.**
14. **Rewrite the following sentence in future tense:**
    خواند.او درس می
15. **Explain the meaning of the idiom:**
    آب از سر گذشتن.

SECTION D: PASHTO

16. **Translate the sentence into Pashto:**
    Education is essential for everyone.
17. **Identify the verb in the following sentence:**
    ما شوم په کورکې لوبې کوي.
18. **Write 4–5 sentences about a Pashto folk story or proverb.**
19. **Convert the following sentence into past tense:**
    زه ښوونځي ته ځم.
20. **Explain the meaning of the Pashto proverb:**
    غروند په رغروند وند ه ریږي.

SECTION E: SINDHI

21. Translate into English:
    ع لم اڌ سان جي زڌ دگي ۾ اهم آهي .
22. Identify the subject and object in:
    ا سڪ تاد شاگ ردن کي سڪ بق پڙهايو.
23. Write a short paragraph about Sindhi literature.
24. Change the verb tense to past:
    مان ا سڪول وڃان ٿو.
25. Explain the meaning of:
    جڪ تي لوڪ، اتي رواج.

SECTION F: PUNJABI

26. Translate into English:
    بڪ پهت ضروری اے ۔ع لم اڌ سان دی زڌ دگی وچ
27. Identify the adjective in:
    لڪ ڙکا ا ساده لڪ باس پڪ پهڪ نے ہوئے ہے ۔
28. Write 5 sentences about Punjabi poetry and its themes.
29. Convert to negative:
    اوه سڪ کول جاڌدا اے ۔
30. Explain the meaning of the proverb:
    جڪ تڪ ھے بڪ چه ہڪسدے اوتڪ ھے گ ھر رو شن ہوڌدا اے ۔

SECTION G: BALOCHI

31. Translate:
    Knowledge lights the mind.
32. Identify the tense in:
    خواڌم.من کڪ تاب می
33. Write 4–5 lines about Balochi folktales.
34. Convert the sentence into future tense:
    او پڪ یڪ شے م یں ا سڪ تاد ا ست.
35. Explain the meaning of a Balochi proverb.

SECTION H: HINDI

36. Translate into Hindi:
    Happiness comes from helping others.
37. Identify the verb:
    बच्चा पार्क में खेल रहा है।
38. Write a short paragraph (5–6 lines) about a famous Hindi author.
39. Rewrite in past tense:
    मैं कल स्कूल जाता हूँ।
40. Explain the meaning of:
    जैसा बोओगे, वैसा काटोगे।

SECTION I: BENGALI

41. Translate into Bengali:
    Education is a powerful tool.
42. Identify the noun in:
    ছেলে বাগানে খেলছে।
43. Write 4–5 sentences about Rabindranath Tagore's poetry.
44. Convert into negative:
    সে স্কুলে যায়।
45. Explain the meaning of the Bengali proverb:
    কাজের মধ্যে আনন্দ আছে।

SECTION J: TAMIL

46. Translate into Tamil:
    Knowledge is light.
47. Identify the verb in:
    மாணவன் பள்ளிக்குச் செல்கிறான்.
48. Write 4–5 lines about Tamil classical literature.
49. Rewrite in past tense:
    நான் புத்தகம் படிக்கிறேன்.
50. Explain the meaning of a Tamil proverb.

SECTION K: TELUGU

51. **Translate into Telugu:**
    Learning never ends.
52. **Identify the subject and object:**
    వీల్లాడు పున్తకన్ని చదువుతున్నడు.
53. **Write a short paragraph on Telugu poetry.**
54. **Convert into future tense:**
    నేను పరశలకు వెళ్తున్నను.
55. **Explain the meaning of a 'Telugu proverb.**

SECTION L: MALAYALAM

56. **Translate into Malayalam:**
    Knowledge empowers people.
57. **Identify the tense:**
    കുട്ടി സ്കൂളിലക്ക് പോകുന്നു.
58. **Write 4–5 lines about Malayalam literature.**
59. **Convert the sentence to negative:**
    അവൻ പഠിക്കുന്നു.
60. **Explain the meaning of a Malayalam proverb.**

SECTION M: SINHALA

61. **Translate into Sinhala:**
    Education is the key to success.
62. **Identify the verb:**
    ළමයා පාසලට යයි.
63. **Write a short paragraph on Sinhala folk tales.**
64. **Convert to past tense:**
    මම පාසලට යනවා.
65. **Explain the meaning of a Sinhala proverb.**

SECTION N: NEPALI

66. **Translate into Nepali:**
    Knowledge changes lives.

67. **Identify the noun in:**
    बालक खेल मैदानमा खेल्दैछ।
68. **Write 4–5 sentences on Nepali literature.**
69. **Convert into future tense:**
    म विद्यालय जाँदैछु।
70. **Explain the meaning of a Nepali proverb.**

SECTION O: CHINESE (MANDARIN)

71. **Translate into Mandarin:**
    Learning is a lifelong journey.
72. **Identify the subject and verb:**
    学生在教室里读书。
73. **Write 3–4 sentences about Chinese classical poetry.**
74. **Rewrite in past tense:**
    我每天学习中文。
75. **Explain the meaning of the proverb:**
    读万卷书，行万里路。

SECTION P: JAPANESE

76. **Translate into Japanese:**
    Knowledge is power.
77. **Identify the verb:**
    学生は公園で遊んでいます。
78. **Write 4–5 lines about a Japanese author or poet.**
79. **Convert into negative form:**
    私は学校に行きます。
80. **Explain the meaning of a Japanese proverb.**

SECTION Q: KOREAN

81. **Translate into Korean:**
    Education shapes the future.
82. **Identify the subject and predicate:**
    학생이 도서관에서 책을 읽고 있다.

83. **Write 3–4 sentences about Korean literature.**
84. **Rewrite in past tense:**
    나는 매일 공부한다.
85. **Explain a Korean proverb.**

SECTION R: THAI

86. **Translate into Thai:**
    Knowledge opens doors.
87. **Identify the verb:**
    เด็กกำลังเล่นในสวน.
88. **Write 3–4 sentences about Thai poetry.**
89. **Convert into future tense:**
    ฉันไปโรงเรียนทุกวัน.
90. **Explain the meaning of a Thai proverb.**

Name: _____

Grade / Class: _____

COMPREHENSIVE EUROPEAN & AFRICAN/INDIGENOUS LANGUAGES TEST

(French, German, Spanish, Italian, Portuguese, Russian, Greek, Latin, Swahili, Hausa, Yoruba, Zulu, Indigenous / Regional Languages)

SECTION A: FRENCH

1. **Translate into French:**
   Knowledge is the foundation of success.
2. **Identify the verb in:**
   Les étudiants lisent des livres à la bibliothèque.
3. **Write a 4–5 sentence paragraph about French literature.**
4. **Rewrite the sentence in future tense:**
   Je mange une pomme.

5. **Explain the meaning of the French proverb:**
   Qui ne risque rien n'a rien.

SECTION B: GERMAN

6. **Translate into English:**
   Bildung öffnet Türen.
7. **Identify the subject and object in:**
   Der Lehrer erklärt den Schülern die Aufgabe.
8. **Write a short paragraph on a famous German author.**
9. **Convert into past tense:**
   Ich gehe jeden Tag zur Schule.
10. **Explain the German idiom:**
    Übung macht den Meister.

SECTION C: SPANISH

11. **Translate into Spanish:**
    Reading develops the mind.
12. **Identify the adjective:**
    La niña pequeña juega en el parque.
13. **Write 4–5 sentences about Spanish poetry.**
14. **Rewrite in negative:**
    Él estudia todos los días.
15. **Explain the meaning of the proverb:**
    Más vale tarde que nunca.

SECTION D: ITALIAN

16. **Translate into Italian:**
    Knowledge is light.
17. **Identify the verb in:**
    Lo studente legge un libro.
18. **Write a short paragraph about a famous Italian poet.**
19. **Convert into past tense:**
    Vado a scuola ogni mattina.

20. **Explain the Italian proverb:**
    Chi dorme non piglia pesci.

SECTION E: PORTUGUESE

21. **Translate into Portuguese:**
    Education shapes the future.
22. **Identify the subject and predicate:**
    O aluno estuda na biblioteca.
23. **Write 4–5 sentences about Portuguese literature.**
24. **Convert into future tense:**
    Eu leio um livro.
25. **Explain the meaning of the proverb:**
    Quem cedo madruga, Deus ajuda.

SECTION F: RUSSIAN

26. **Translate into English:**
    Знание – сила.
27. **Identify the noun in:**
    Мальчик играет в саду.
28. **Write a short paragraph about a famous Russian author.**
29. **Convert into past tense:**
    Я учусь каждый день.
30. **Explain the meaning of the Russian proverb:**
    Без труда не вытащишь и рыбку из пруда.

SECTION G: GREEK

31. **Translate into Greek:**
    Education opens the mind.
32. **Identify the verb:**
    Ο μαθητής διαβάζει ένα βιβλίο.
33. **Write 4–5 sentences on Greek mythology or literature.**
34. **Rewrite in future tense:**
    Πηγαίνω στο σχολείο κάθε μέρα.

35. **Explain the meaning of the Greek proverb:**
    Γνῶθι σεαυτόν.

SECTION H: LATIN

36. **Translate into Latin:**
    Knowledge is power.
37. **Identify the subject and object in:**
    Puella librum legit in horto.
38. **Write 4–5 lines about a Roman author or philosopher.**
39. **Convert into past tense:**
    Puer ad scholam it.
40. **Explain the Latin proverb:**
    Carpe diem.

SECTION I: SWAHILI

41. **Translate into Swahili:**
    Learning is a lifelong journey.
42. **Identify the verb:**
    Mwanafunzi anasoma kitabu.
43. **Write 4–5 sentences about Swahili poetry or folktales.**
44. **Convert into negative:**
    Mimi naenda shule kila siku.
45. **Explain the Swahili proverb:**
    Haraka haraka haina baraka.

SECTION J: HAUSA

46. **Translate into Hausa:**
    Education changes lives.
47. **Identify the noun in:**
    Yaro yana wasa a filin wasa.
48. **Write a short paragraph about Hausa literature.**
49. **Convert into past tense:**
    Ina zuwa makaranta.

50. **Explain the meaning of the Hausa proverb:**
    Komai nisan jifa ƙasa zai dawo.

SECTION K: YORUBA

51. **Translate into Yoruba:**
    Knowledge opens doors.
52. **Identify the subject and predicate:**
    Ọmọde naa nṣere ni papa.
53. **Write 4–5 sentences about Yoruba oral literature.**
54. **Rewrite in future tense:**
    Mo n lọ si ile-iwe lojoojumọ.
55. **Explain the Yoruba proverb:**
    Bi a ko ba gbin ogede, ko le gbe eso.

SECTION L: ZULU

56. **Translate into Zulu:**
    Education empowers everyone.
57. **Identify the verb:**
    Umfundi udlala epaki.
58. **Write 4–5 lines about Zulu folktales or literature.**
59. **Convert into negative:**
    Ngiyafunda nsuku zonke.
60. **Explain the Zulu proverb:**
    Isandla sihlamba esinye.

SECTION M: INDIGENOUS / REGIONAL LANGUAGES

61. **Choose an Indigenous/Regional language you know and translate:**
    Education is the key to success.
62. **Identify the subject, verb, and object in a sentence of that language.**
63. **Write 4–5 sentences about a famous folk story, poem, or proverb in that language.**
64. **Convert a present tense sentence to past tense.**
65. **Explain a proverb from that Indigenous or regional language.**

Name: _____

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MATHEMATICS & LOGICAL SCIENCES – COMPREHENSIVE TEST

SECTION A: GENERAL MATHEMATICS

1. Solve for x: 7x−9=2x+167x - 9 = 2x + 167x−9=2x+16.
2. A cylindrical tank has a radius of 5 m and a height of 10 m. Find the volume and surface area.
3. Simplify: 4(3x−5)−2(2x+7)4(3x - 5) - 2(2x + 7)4(3x−5)−2(2x+7).
4. Solve the quadratic equation: x2−11x+28=0x^2 - 11x + 28 = 0x2−11x+28=0.
5. If the sum of three consecutive integers is 78, find the integers.
6. Convert 135 degrees into radians.
7. A shop sold 120 pencils at $2 each and 80 pens at $3 each. Find the total revenue.
8. Factorize: x2−16x^2 - 16x2−16.
9. Solve for y: 3(y−2)+5=2(y+4)3(y - 2) + 5 = 2(y + 4)3(y−2)+5=2(y+4).
10. A train covers 150 km in 3 hours. Calculate its average speed.

SECTION B: PURE MATHEMATICS

11. Prove: (a−b)2=a2−2ab+b2(a - b)^2 = a^2 - 2ab + b^2(a−b)2=a2−2ab+b2.
12. Solve for x: 2x2−5x−3=02x^2 - 5x - 3 = 02x2−5x−3=0 using factorization.
13. Find the sum of the first 50 odd numbers.
14. Solve: 3cos⁡θ−2=03\cos \theta - 2 = 03cosθ−2=0 for 0≤θ≤360∘0 \leq \theta \leq 360^\circ0≤θ≤360∘.
15. If log⁡2x=5\log_2 x = 5log2x=5, find x.
16. Find the derivative of f(x)=3x4−5x2+2x−7f(x) = 3x^4 - 5x^2 + 2x - 7f(x)=3x4−5x2+2x−7.
17. Simplify: 2x2−84x\frac{2x^2 - 8}{4x}4x2x2−8.
18. If x+y=10x + y = 10x+y=10 and xy=21xy = 21xy=21, find the values of x and y.
19. Solve the system of equations:

{x+2y=83x−y=5\begin{cases} x + 2y = 8 \\ 3x - y = 5 \end{cases}{x+2y=83x−y=5

20. Prove that the sum of angles in a triangle is 180°.

SECTION C: APPLIED MATHEMATICS

21. A car depreciates 15% per year. If its current value is $20,000, find its value after 3 years.
22. Calculate the simple interest on $5,000 at 8% per annum for 3 years.
23. A factory produces 500 units per day. How many units are produced in 45 days?
24. A machine costs $10,000 and its salvage value is $2,000 after 5 years. Calculate the straight-line depreciation per year.
25. A population grows at a rate of 2% per year. If the current population is 50,000, find the population after 10 years.
26. Solve: 5x+7y=355x + 7y = 355x+7y=35, if y=3y = 3y=3.
27. Calculate the area of a triangle with base 12 cm and height 7 cm.
28. A sum of money doubles in 8 years at simple interest. Find the rate of interest.
29. A train travels 60 km in the first hour and 90 km in the second hour. Find its average speed.
30. Solve for x: 2x3+54=72\frac{2x}{3} + \frac{5}{4} = \frac{7}{2}32x+45=27.

SECTION D: ADVANCED MATHEMATICS

31. Solve: ddx(5x3−2x2+4x−7)\frac{d}{dx} (5x^3 - 2x^2 + 4x - 7)dxd(5x3−2x2+4x−7).
32. Integrate: ∫(3x2−4x+5)dx\int (3x^2 - 4x + 5) dx∫(3x2−4x+5)dx.
33. Find limx→2x2−4x−2\lim_{x \to 2} \frac{x^2 - 4}{x - 2}limx→2x−2x2−4.
34. Solve the differential equation: dydx=3x2\frac{dy}{dx} = 3x^2dxdy=3x2.
35. Find the second derivative of f(x)=x4−3x3+2xf(x) = x^4 - 3x^3 + 2xf(x)=x4−3x3+2x.
36. Evaluate: ∫05(2x+3)dx\int_0^5 (2x + 3) dx∫05(2x+3)dx.
37. Determine the maxima and minima of f(x)=x3−6x2+9x+2f(x) = x^3 - 6x^2 + 9x + 2f(x)=x3−6x2+9x+2.
38. Solve: d2ydx2−5dydx+6y=0\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0dx2d2y−5dxdy +6y=0.
39. If f(x)=lnxf(x) = \ln xf(x)=lnx, find f′(x)f'(x)f′(x) and f″(x)f''(x)f″(x).
40. Find the partial derivative ∂∂x(x2y+3xy2)\frac{\partial}{\partial x} (x^2y + 3xy^2)∂x∂ (x2y+3xy2).

SECTION E: STATISTICS

41. **Find the mean of: 4, 7, 12, 15, 20.**
42. **Calculate the median of: 8, 5, 12, 9, 15.**
43. **Find the mode of: 3, 3, 5, 7, 7, 7, 9.**
44. **Compute the standard deviation for: 5, 7, 9, 11, 13.**
45. **A die is rolled 50 times. The results of 6 appear 8 times. Find the experimental probability.**
46. **Draw a frequency table for the marks: 12, 15, 18, 12, 20, 18, 12.**
47. **Calculate the variance of: 2, 4, 6, 8, 10.**
48. **Construct a cumulative frequency table for: 1, 2, 2, 3, 4, 4, 5.**
49. **Explain the difference between population and sample in statistics.**
50. **Determine the quartiles for: 3, 7, 8, 5, 12, 14, 18, 20.**

SECTION F: PROBABILITY

51. **A coin is tossed 3 times. Find the probability of getting exactly 2 heads.**
52. **A bag contains 3 red, 5 green, and 2 blue balls. One ball is drawn at random. Find the probability it is green.**
53. **Two dice are rolled. Find the probability the sum is 7.**
54. **From a deck of 52 cards, find the probability of drawing an Ace or a King.**
55. **A student passes Math with probability 0.8 and English with probability 0.7. Assuming independence, find the probability the student passes both subjects.**
56. **Explain mutually exclusive events with an example.**
57. **A box contains 6 white and 4 black balls. Two balls are drawn without replacement. Find the probability both are black.**
58. **A coin is tossed until the first head appears. What is the probability it appears on the 3rd toss?**
59. **If events A and B are independent, show that P(A∩B)=P(A)×P(B)P(A \cap B) = P(A) \times P(B)P(A∩B)=P(A)×P(B).**
60. **A bag has 5 red and 7 green balls. Find the probability of drawing at least one red in two draws with replacement.**

SECTION G: CALCULUS (Differential, Integral, Multivariable)

61. Differentiate: $f(x) = x^3 - 5x + 7$.

62. Integrate: $\int (6x^2 + 4x - 3)dx$.

63. Evaluate: $\int_1^3 (2x + 5)dx$.

64. Find $\frac{\partial}{\partial y}(x^2y + 4xy^2)$.

65. Find the derivative of $y = \sin x \cdot \cos x$.

66. Solve: $\frac{dy}{dx} = 4x^3 - 2x$.

67. Evaluate $\lim_{x \to 0} \frac{\sin 3x}{x}$.

68. Find the gradient of $f(x, y) = x^2 + y^2 + xy$ at point (1,2).

69. Integrate: $\int xe^x dx$.

70. Solve $\frac{\partial^2 f}{\partial x^2} = 12x^2$ if $f(0) = 0$.

71. Solve the system using matrices:

$$\begin{cases} x + 2y = 5 \\ 3x - y = 4 \end{cases}$$

72. Find the determinant of:

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

73. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, find $A + B$ and AB.

74. Find the inverse of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

75. Determine whether the vectors $\vec{v_1} = (1, 2, 3)$ and $\vec{v_2} = (2, 4, 6)$ are linearly independent.

76. Find the eigenvalues of $\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.

77. Solve: $A\vec{x} = \vec{b}$, where $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$.

78. Compute the rank of: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

79. Determine if the matrix $\begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}$ is invertible.

80. Solve $\vec{x}^T A\vec{x}$ for $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

SECTION I: DISCRETE MATHEMATICS

81. Determine if the graph with vertices {A, B, C, D} and edges {AB, AC, BD} is connected.

82. Find the number of subsets of the set {1,2,3,4,5}.

83. Construct the truth table for $P \wedge (Q \vee R)$.

84. Use mathematical induction to prove that $1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}$.

85. Find the GCD of 84 and 120 using Euclidean algorithm.

86. Determine whether the function $f : \mathbb{Z} \to \mathbb{Z}, f(x) = 2x + 3$ is injective, surjective, or bijective.

87. Solve the recurrence relation $a_n = 3a_{n-1} - 2$ with $a_1 = 5$.

88. Represent the number 29 in binary.

89. Find the number of arrangements of the letters in the word "ALGEBRA".

90. Determine if the following relation on {1,2,3} is reflexive, symmetric, and transitive: $R = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$.

SECTION J: NUMERICAL METHODS

91. Use the bisection method to approximate the root of $f(x) = x^3 - x - 2$ in [1,2] up to 2 decimal places.

92. Apply the Newton-Raphson method to find the root of $x^2 - 5 = 0$ starting with $x_0 = 2$.

93. Solve $2x + y = 5, x - y = 1$ using Gauss-Seidel iteration method.

94. Approximate the derivative of $f(x) = x^2$ at $x = 3$ using forward difference with $h = 0.01$.

95. Use Simpson's 1/3 rule to estimate $\int_0^2 (x^2 + 1)dx$ with 4 subintervals.

96. Solve the linear system using Jacobi iteration:

$$\begin{cases} 10x - y + 2z = 6 \\ -2x + 8y - z = -5 \\ x + 3y + 12z = 15 \end{cases}$$

97. Find the approximate solution of $y' = x + y$ using Euler's method with step size h = 0.1, initial condition y(0) = 1, at x = 0.1.

98. Determine the largest eigenvalue of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ using the power method.

99. Solve $f(x) = e^x - 2 = 0$ using one iteration of fixed-point iteration starting from $x_0 = 0$.

100. Use Lagrange interpolation to find the polynomial passing through points (0,1), (1,3), (2,7).

SECTION K: MATHEMATICAL MODELING

101. A tank is filled at 10 liters per minute and emptied at 8 liters per minute. Model the water level as a function of time.

102. A population of rabbits grows according to $P(t) = P_0 e^{0.03t}$. Find the population after 5 years if $P_0 = 200$.

103. Construct a linear model for cost C and units produced x, given: 100 units cost $500, 200 units cost $900.

104. A company has two machines producing items at different rates. Model the combined production function.

105. Formulate a mathematical model for traffic flow assuming the number of cars entering a road per hour is proportional to the time of day.

106. Use matrices to model a system of 3 interconnected banks with money flow between them.

107. Model the decay of a radioactive substance with half-life 10 hours.

108. Construct a quadratic model to fit data points: (1,3), (2,5), (3,10).

109. Model the trajectory of a projectile launched at 20 m/s at 30° angle from horizontal.

110. Use a system of linear inequalities to model a factory's production constraints for 2 products.

111. Solve the following linear programming problem using graphical method: Maximize $Z = 3x + 2y$ subject to $x + y \leq 4, x \geq 0, y \geq 0$.

112. Apply the Simplex method to maximize $Z = 2x + 5y$ given constraints:

$$x + 2y \leq 8, 3x + y \leq 9, x, y \geq 0$$

113. Use transportation method to minimize cost for transporting goods from 2 factories to 3 warehouses with given cost table.

114. Solve the assignment problem for 3 workers and 3 jobs with cost matrix:

$$\begin{bmatrix} 4 & 2 & 5 \\ 3 & 7 & 6 \\ 8 & 1 & 4 \end{bmatrix}$$

115. Apply critical path method (CPM) to a project with 5 activities and given durations.

116. Determine the expected project completion time using PERT for a 3-activity project.

117. Formulate a production scheduling problem as a linear programming model.

118. Use queuing theory to determine average waiting time in a single-server system with arrival rate λ=5 and service rate μ=8.

119. Solve the inventory management problem using EOQ formula with demand 1000 units/year, ordering cost $50, holding cost $2/unit.

120. Apply sensitivity analysis on a linear programming problem to determine effect of increasing a constraint.

Name: _____

Grade/Class: 10–16

Subject: Natural & Physical Sciences Comprehensive Test

SECTION A: PHYSICS

Classical Physics

1. **A ball is thrown vertically upwards with an initial speed of 20 m/s. Calculate the maximum height it reaches.**
2. **Determine the acceleration of an object sliding down a frictionless incline of angle 30° and height 5 m.**
3. **A car accelerates from rest to 30 m/s in 10 seconds. Find its acceleration and distance traveled.**
4. **Calculate the work done in moving a 50 kg object through a vertical height of 10 m.**
5. **A pendulum has a length of 2 m. Find its time period.**

Modern Physics

6. **Explain the photoelectric effect and determine the stopping potential if the work function is 2 eV and incident light has a frequency of 1.5×10151.5 \times 10^{15}1.5×1015 Hz.**
7. **An electron is accelerated through a potential difference of 100 V. Calculate its final speed.**
8. **Compare Newtonian mechanics and relativistic mechanics for a particle moving at 0.8c.**

Quantum Physics (Intro)

9. **A photon has a wavelength of 500 nm. Calculate its energy.**
10. **State Heisenberg's uncertainty principle and calculate the uncertainty in velocity if a particle's position is known within 1 nm.**

Nuclear Physics (Intro)

11. **The half-life of a radioactive isotope is 4 hours. If the initial mass is 16 g, how much remains after 12 hours?**
12. **Describe the difference between alpha, beta, and gamma radiation.**
13. **Calculate the binding energy of a nucleus if mass defect is 0.005 u.**

Astrophysics & Space Science

14. **Calculate the escape velocity from Earth's surface.**
15. **A star has a luminosity of 3.8×10263.8 \times 10^{26}3.8×1026 W and radius 7×1087 \times 10^87×108 m. Calculate its surface temperature using Stefan-Boltzmann law.**
16. **Explain the Doppler effect in the context of light emitted from a receding galaxy.**

Geophysics

17. **Explain how seismic waves are used to determine Earth's interior structure.**
18. **Calculate the gravitational potential energy of a 10 kg mass at a depth of 500 m inside the Earth, assuming ggg is uniform.**

Applied & Electronics Physics

19. **A circuit has a 12 V battery connected in series with a 4 Ω resistor and a 6 Ω resistor. Find the current and voltage drop across each resistor.**
20. **Determine the equivalent capacitance of three capacitors 2 µF, 3 µF, and 6 µF in series.**
21. **Describe how semiconductors are used in diodes and transistors.**

SECTION B: CHEMISTRY

General Chemistry

22. **Balance the chemical reaction: C2H6+O2→CO2+H2O\text{C}_2\text{H}_6 + O_2 \rightarrow CO_2 + H_2OC2H6+O2→CO2+H2O.**
23. **Calculate the molar mass of Na2SO4Na_2SO_4Na2SO4.**
24. **Determine the number of moles in 25 g of water.**

Organic Chemistry

25. **Draw the structural formula of butane and explain its isomers.**
26. **Name the following compound: CH3CH2OHCH_3CH_2OHCH3CH2OH.**

27. **Explain the mechanism of free radical halogenation of methane.**

Inorganic Chemistry

28. **Write the electronic configuration of Iron (Fe).**
29. **Explain the difference between ionic and covalent bonds with examples.**
30. **Predict the product of reaction between AlAlAl and HClHClHCl.**

Physical Chemistry

31. **Using the ideal gas law, calculate the volume of 1 mole of gas at 2 atm pressure and 300 K.**
32. **Explain Hess's law and calculate enthalpy change for C+O2→CO2C + O_2 \rightarrow CO_2C+O2→CO2.**

Analytical & Industrial Chemistry

33. **Describe the titration process to determine the concentration of HCl using NaOH.**
34. **Explain the industrial production of ammonia by Haber process.**

Environmental & Pharmaceutical Chemistry

35. **Describe the process of water purification in municipal plants.**
36. **Explain how aspirin is synthesized and its chemical structure.**

Polymer & Nanochemistry

37. **Name three types of polymers and their uses.**
38. **Explain how nanoparticles are used in drug delivery systems.**

SECTION C: BIOLOGY

General Biology

39. **Explain the difference between prokaryotic and eukaryotic cells.**
40. **Draw and label a diagram of an animal cell.**

Botany

41. **Explain photosynthesis and write the chemical equation.**
42. **Describe the structure and function of xylem and phloem.**

Zoology

43. **Explain the process of digestion in humans.**
44. **Describe the structure of a mammalian heart and blood flow.**

Microbiology

45. **Name the main types of bacteria and describe their shapes.**
46. **Explain the role of microorganisms in nitrogen fixation.**

Genetics & Molecular Biology

47. **Explain Mendel's laws of inheritance with examples.**
48. **Describe the structure of DNA and its replication process.**

Biotechnology & Biochemistry

49. **Explain the process of PCR (Polymerase Chain Reaction).**
50. **Identify the three main macromolecules in living organisms and their functions.**

Human & Marine Biology

51. **Describe the human respiratory system.**
52. **Explain adaptations of fish to aquatic environment.**

Environmental & Agricultural Biology

53. **Explain the role of decomposers in ecosystems.**
54. **Describe crop rotation and its benefits in agriculture.**

Biomedical Sciences

55. **Describe the mechanism of action of vaccines.**
56. **Explain the basic principles of blood transfusion compatibility.**

Name: _____

Grade/Class: 11–16

Subject: Medical & Health Sciences Comprehensive Test

SECTION A: PRE-MEDICAL STUDIES & HUMAN ANATOMY

1. **Describe the structure of a typical human cell and explain the function of at least five organelles.**
2. **Draw a labeled diagram of the human skeletal system and identify the bones of the upper limb.**
3. **Explain the differences between compact bone and spongy bone.**
4. **Describe the structure and function of the human heart, including the names of chambers, valves, and major blood vessels.**
5. **Identify and explain the functions of the major human organ systems: circulatory, respiratory, digestive, excretory, nervous, and endocrine systems.**
6. **Explain the anatomical differences between arteries, veins, and capillaries.**
7. **Describe the human respiratory system, including the pathway of air from the nose to alveoli.**
8. **Explain the structure and function of the human kidney and nephron.**
9. **Describe the structure of the human brain and its major regions, along with their functions.**
10. **Compare and contrast male and female reproductive systems.**

SECTION B: HUMAN PHYSIOLOGY

11. **Explain the cardiac cycle, including systole and diastole.**
12. **Describe the mechanism of breathing and the role of diaphragm and intercostal muscles.**
13. **Explain how oxygen and carbon dioxide are transported in the blood.**
14. **Describe the process of digestion, absorption, and assimilation of nutrients in humans.**
15. **Explain the process of nerve impulse transmission through neurons.**
16. **Describe the hormonal regulation of blood glucose levels by insulin and glucagon.**
17. **Explain the role of the liver in metabolism and detoxification.**
18. **Describe the process of thermoregulation in humans.**
19. **Explain how the immune system responds to pathogens.**
20. **Discuss the physiological changes during exercise on the cardiovascular and respiratory systems.**

SECTION C: PATHOLOGY & PHARMACOLOGY (INTRO)

21. **Define pathology and explain its significance in medicine.**
22. **Describe the differences between acute and chronic diseases.**
23. **Explain the basic mechanisms of inflammation and its signs.**
24. **Define pharmacology and describe the differences between pharmacokinetics and pharmacodynamics.**
25. **List and explain the different routes of drug administration.**
26. **Explain the action and side effects of at least two common antibiotics.**
27. **Discuss the difference between generic and brand drugs.**
28. **Explain how vaccines work to prevent infectious diseases.**
29. **Identify common cardiovascular and respiratory disorders and their basic symptoms.**
30. **Describe the impact of lifestyle and environmental factors on disease development.**

SECTION D: NUTRITION, PUBLIC & COMMUNITY HEALTH

31. **Describe the functions of carbohydrates, proteins, and fats in the human diet.**
32. **Explain the importance of vitamins and minerals and give examples of deficiencies.**
33. **Describe the nutritional requirements for different age groups.**
34. **Explain the concept of balanced diet and meal planning.**
35. **Discuss the role of water and electrolytes in maintaining homeostasis.**
36. **Explain the principles of public health and its importance in disease prevention.**
37. **Describe common public health interventions for controlling infectious diseases.**
38. **Explain the role of community health workers in promoting health awareness.**
39. **Discuss methods for preventing malnutrition in developing countries.**
40. **Describe the importance of sanitation and hygiene in community health.**

SECTION E: MEDICAL ETHICS, FIRST AID & EMERGENCY CARE

41. **Explain the principles of medical ethics, including autonomy, beneficence, non-maleficence, and justice.**
42. **Describe patient confidentiality and informed consent.**
43. **Explain the steps in providing first aid for burns, cuts, and fractures.**
44. **Describe the procedure for cardiopulmonary resuscitation (CPR).**
45. **Explain the management of choking in adults and children.**
46. **Describe the principles of triage in emergency care situations.**
47. **Explain the use of automated external defibrillators (AEDs) in cardiac emergencies.**

48. Describe the first aid procedures for poisoning and allergic reactions.
49. Explain the importance of infection control during first aid.
50. Discuss ethical dilemmas that may arise during emergency care.

SECTION F: NURSING STUDIES & PHYSIOTHERAPY (INTRO)

51. Describe the basic roles and responsibilities of a nurse in patient care.
52. Explain the importance of hygiene and sterilization in nursing practice.
53. Discuss techniques for patient mobility and safe lifting practices.
54. Explain the role of physiotherapy in rehabilitation.
55. Describe common physiotherapy exercises for improving joint mobility.
56. Explain the benefits of physiotherapy in post-operative recovery.
57. Discuss the principles of pain management in nursing and physiotherapy.
58. Describe methods to prevent bedsores in immobile patients.
59. Explain the importance of patient education in nursing and physiotherapy.
60. Describe how physiotherapy can assist patients with respiratory disorders.

SECTION G: SPORTS MEDICINE, MENTAL HEALTH & PSYCHOLOGY (CLINICAL INTRO)

61. Explain the importance of warm-up and cool-down in exercise.
62. Describe common sports injuries and their basic management.
63. Discuss the role of nutrition and hydration in sports performance.
64. Explain the effects of exercise on cardiovascular and muscular systems.
65. Describe common mental health disorders, such as depression and anxiety.
66. Explain the basic principles of cognitive-behavioral therapy (CBT).
67. Discuss the importance of stress management for mental well-being.
68. Explain the role of psychologists in clinical and community settings.
69. Describe the effects of sleep on mental health and cognitive performance.
70. Discuss strategies for promoting psychological resilience in patients.

5 ENGINEERING & TECHNOLOGY

Core Engineering

- Civil Engineering (Intro)
- Mechanical Engineering (Intro)
- Electrical Engineering (Intro)

- **Electronics Engineering**
- **Chemical Engineering**
- **Environmental Engineering**
- **Industrial Engineering**
- **Mechatronics**
- **Robotics Engineering**

Technology

- **Computer Science**
- **Software Engineering**
- **Information Technology (IT)**
- **Artificial Intelligence**
- **Machine Learning**
- **Data Science**
- **Cloud Computing**
- **Cybersecurity**
- **Blockchain Technology**
- **Internet of Things (IoT)**
- **Game Development**
- **Mobile App Development**

⍰ Computer Applications & Office Skills Test

Instructions:

- **Attempt all questions.**
- **For practical questions, describe your steps clearly.**
- **Where applicable, provide screenshots, tables, or diagrams.**
- **Covers Digital Literacy, Computer Applications, Programming, Web, Database, Networking, OS, MS Office, Google Workspace, Typing, Data Entry, and Office Automation.**

Section A: Digital Literacy (5 Questions)

1. **Define digital literacy and explain its importance in the modern workplace.**
2. **Explain the difference between hardware and software.**
3. **What are examples of safe online practices for data protection?**
4. **Describe the basic functions of input, output, and storage devices.**
5. **Explain the difference between local storage and cloud storage.**

Section B: Computer Applications (5 Questions)

1. **List three common computer applications used in offices and their purposes.**
2. **Explain the difference between system software and application software.**
3. **Describe multimedia applications and their use in presentations.**
4. **Explain the role of office productivity suites.**
5. **Discuss how software updates improve performance and security.**

Section C: Programming Fundamentals (5 Questions)

1. **Define programming and explain its basic concepts (variables, loops, conditions).**
2. **Write a pseudocode to calculate the sum of the first 50 natural numbers.**
3. **Explain the difference between compiled and interpreted languages.**
4. **Describe the use of functions and procedures in programming.**
5. **Explain debugging and its importance in programming.**

Section D: Web Design (5 Questions)

1. **Define web design and explain its importance.**
2. **What is the difference between UI and UX design?**
3. **List 5 essential HTML tags and their functions.**
4. **Explain the concept of responsive design.**
5. **Discuss color schemes and typography in website design.**

Section E: Web Development (5 Questions)

1. **Explain the difference between front-end and back-end development.**
2. **What is the role of CSS and JavaScript in web development?**
3. **Describe the concept of a web server and client.**
4. **Explain how forms and input fields are used to collect user data.**
5. **Discuss the importance of website testing and debugging.**

Section F: Database Management (5 Questions)

1. **Define a database and a database management system (DBMS).**
2. **Explain the difference between a primary key and a foreign key.**
3. **What are the advantages of using a relational database?**
4. **Describe SQL and write a query to select all records from a table named "Employees".**
5. **Explain data backup and recovery in database systems.**

Section G: Networking (5 Questions)

1. **Define computer network and explain types: LAN, WAN, MAN.**
2. **Explain the role of routers and switches in a network.**
3. **Describe the difference between IP address and MAC address.**
4. **Explain basic network security practices.**
5. **What is the difference between wired and wireless networking?**

Section H: Operating Systems (5 Questions)

1. **Define an operating system and its main functions.**
2. **Compare Windows, Linux, and Mac OS.**
3. **Explain multitasking and memory management in operating systems.**
4. **What is a file system? Give examples.**
5. **Explain how user accounts and permissions are managed in OS.**

Section I: Office & Productivity (5 Questions)

1. **Define office automation and its benefits in business.**
2. **List 5 key productivity tools used in offices.**
3. **Explain the difference between collaboration tools and communication tools.**
4. **Describe the role of document management systems.**
5. **Discuss how productivity software improves workflow efficiency.**

Section J: MS Word (Beginner → Advanced) (5 Questions)

1. **How do you create a table of contents in MS Word?**
2. **Explain the difference between "Track Changes" and "Comments".**
3. **How do you insert headers, footers, and page numbers?**
4. **Describe how to use mail merge for letters or emails.**
5. **Explain advanced formatting techniques: styles, themes, and templates.**

Section K: MS Excel (Beginner → Advanced) (5 Questions)

1. **Write a formula to calculate the sum of cells A1 to A10.**
2. **Explain the difference between absolute, relative, and mixed cell references.**
3. **Describe how to create charts and graphs from data.**
4. **Explain how to use VLOOKUP or HLOOKUP functions.**
5. **Discuss pivot tables and their benefits in data analysis.**

Section L: MS PowerPoint (5 Questions)

1. **Explain how to create a professional presentation in MS PowerPoint.**
2. **Describe the use of animations and transitions.**
3. **How do you insert multimedia elements (images, audio, video) in slides?**
4. **Explain slide master and its importance in consistent design.**
5. **Discuss tips for effective presentation delivery.**

Section M: MS Access (5 Questions)

1. **Define MS Access and its key features.**
2. **Explain the difference between a table, query, form, and report.**
3. **Describe how to create relationships between tables.**
4. **Write a query to retrieve all employees with salary > $5000.**
5. **Discuss the benefits of using a database in small businesses.**

Section N: Google Workspace (5 Questions)

1. **List the main applications included in Google Workspace.**
2. **Explain the use of Google Docs, Sheets, and Slides.**

3. **How do you share files and manage permissions in Google Drive?**
4. **Explain collaboration features like comments and real-time editing.**
5. **Discuss the advantages of cloud-based productivity tools.**

Section O: Office Automation (5 Questions)

1. **Define office automation and provide examples of automated tasks.**
2. **Explain how document scanning and digital filing improve efficiency.**
3. **Describe workflow automation in business processes.**
4. **Discuss the benefits of automated scheduling and reminders.**
5. **Explain the use of integrated office suites for seamless operations.**

Section P: Typing & Documentation (5 Questions)

1. **Explain the importance of typing speed and accuracy.**
2. **Describe proper typing techniques for beginners.**
3. **Explain how to format professional documents for official use.**
4. **What are the key elements of a business report?**
5. **Discuss digital note-taking tools and their advantages.**

Section Q: Data Entry (5 Questions)

1. **Explain the role of data entry in office administration.**
2. **Describe common data entry errors and how to prevent them.**
3. **Discuss the importance of accuracy and speed in data entry tasks.**
4. **Explain the use of software tools to automate data entry.**
5. **Describe how data validation ensures high-quality information.**

Section R: Digital Filing Systems (5 Questions)

1. **Define digital filing system and its advantages over paper filing.**
2. **Explain folder structure design for organized digital storage.**
3. **Describe naming conventions and version control for files.**
4. **Discuss backup strategies for digital documents.**

5. **Explain how cloud storage integrates with digital filing systems.**

⍰ Ultimate Social Sciences Test (College Level)

Instructions:

- **Attempt all questions.**
- **Answer theory questions with examples and explanations.**
- **Use maps, diagrams, or tables where needed.**
- **Covers History, Geography, Political Science, Civics, Sociology, Anthropology, Psychology, Social Work, Population & Gender Studies, Cultural Studies, Peace & Conflict Studies, International & Global Studies, Area Studies.**

Section A: History (World, Regional, Islamic) (7 Questions)

1. **Describe the main causes and effects of World War I.**
2. **Explain the rise and fall of the Roman Empire.**
3. **Discuss the major achievements of the Islamic Golden Age.**
4. **Explain the impact of colonialism on South Asia.**
5. **Describe the French Revolution and its significance in world history.**
6. **Discuss the Mughal Empire's administrative and cultural contributions.**
7. **Explain the causes and effects of the Cold War.**

Section B: Geography (Physical, Human, Economic) (7 Questions)

1. **Explain the differences between physical, human, and economic geography.**
2. **Describe the process of plate tectonics and its impact on earth's surface.**
3. **Explain factors affecting population distribution.**
4. **Discuss urbanization and its socio-economic effects.**
5. **Explain the concept of sustainable development in geography.**
6. **Describe major climatic zones of the world.**
7. **Explain the role of natural resources in economic development.**

Section C: Political Science (7 Questions)

1. **Define political science and its importance.**
2. **Explain the differences between democracy, monarchy, and authoritarian regimes.**
3. **Discuss the role of political parties in modern democracies.**
4. **Explain the concept of separation of powers.**
5. **Discuss the significance of civil rights and liberties.**
6. **Explain the concept of governance and good governance indicators.**
7. **Analyze the role of international organizations like the UN in global politics.**

Section D: Civics (7 Questions)

1. **Explain the fundamental rights and duties of citizens.**
2. **Describe the structure of a federal government.**
3. **Explain the electoral process in a democratic system.**
4. **Discuss the role of local governments in public administration.**
5. **Explain the importance of rule of law in society.**
6. **Discuss civic responsibilities in promoting social harmony.**
7. **Explain the role of civil society organizations in public life.**

Section E: Sociology (7 Questions)

1. **Define sociology and its major areas of study.**
2. **Explain the concepts of culture, norms, and values.**
3. **Discuss the role of social institutions in society.**
4. **Explain the impact of social stratification on communities.**
5. **Describe the difference between primary and secondary social groups.**
6. **Discuss the concept of social change and its drivers.**
7. **Explain the role of sociology in understanding modern social issues.**

Section F: Anthropology (7 Questions)

1. **Define anthropology and its main subfields.**
2. **Explain the importance of cultural anthropology in understanding societies.**
3. **Discuss the study of human evolution and its methods.**
4. **Explain the concept of ethnography and its application.**
5. **Describe the role of archaeology in studying past civilizations.**
6. **Discuss the significance of kinship and social organization in anthropology.**
7. **Explain applied anthropology and its relevance in modern society.**

Section G: Psychology (7 Questions)

1. **Define psychology and its main branches.**
2. **Explain the difference between clinical and developmental psychology.**
3. **Discuss major psychological theories: Freud, Piaget, and Skinner.**
4. **Explain cognitive processes: perception, memory, and learning.**
5. **Describe the role of emotions and motivation in human behavior.**
6. **Explain psychological disorders and their classification.**
7. **Discuss the importance of psychological research in society.**

Section H: Social Work (7 Questions)

1. **Define social work and its objectives.**
2. **Explain the role of social workers in communities.**
3. **Discuss methods of social work: casework, group work, and community work.**
4. **Explain the importance of ethics in social work practice.**
5. **Describe challenges faced by social workers in modern society.**
6. **Discuss social welfare policies and their impact.**
7. **Explain the role of NGOs in social development.**

Section I: Population Studies (7 Questions)

1. **Define population studies and its significance.**
2. **Explain factors affecting population growth.**
3. **Discuss the demographic transition model.**
4. **Explain the social and economic impacts of overpopulation.**
5. **Discuss population policies in developing countries.**
6. **Explain migration and its effects on society.**
7. **Describe methods for population data collection.**

Section J: Gender Studies (7 Questions)

1. **Define gender studies and its importance in social sciences.**
2. **Explain the difference between sex and gender.**

3. **Discuss gender roles and stereotypes in society.**
4. **Explain the impact of gender inequality on development.**
5. **Describe women's empowerment initiatives globally.**
6. **Discuss the role of education in promoting gender equality.**
7. **Explain how gender studies intersect with other social sciences.**

Section K: Cultural Studies (7 Questions)

1. **Define culture and its components.**
2. **Explain the role of language, religion, and traditions in culture.**
3. **Discuss cultural diversity and its importance.**
4. **Explain cultural diffusion and globalization effects.**
5. **Describe the impact of mass media on cultural identity.**
6. **Discuss the role of festivals and rituals in society.**
7. **Explain cultural heritage preservation methods.**

Section L: Peace & Conflict Studies (7 Questions)

1. **Define peace studies and conflict studies.**
2. **Explain the types of conflicts: interpersonal, communal, international.**
3. **Discuss conflict resolution strategies and negotiation techniques.**
4. **Explain the role of diplomacy in maintaining peace.**
5. **Discuss the impact of war on societies.**
6. **Describe peace-building initiatives by international organizations.**
7. **Explain the importance of human rights in conflict prevention.**

Section M: International Relations (7 Questions)

1. **Define international relations and its scope.**
2. **Explain theories of international relations: realism, liberalism, constructivism.**
3. **Discuss the role of the United Nations in global governance.**
4. **Explain diplomacy and foreign policy formulation.**
5. **Discuss international trade and economic cooperation.**
6. **Explain the role of treaties and international law.**
7. **Analyze a recent international conflict and its resolution efforts.**

Section N: Global Studies (7 Questions)

1. **Define global studies and its interdisciplinary approach.**
2. **Explain globalization and its effects on economies and cultures.**
3. **Discuss global challenges: climate change, poverty, and migration.**
4. **Explain the role of international organizations in addressing global issues.**
5. **Discuss the impact of global media on society.**
6. **Describe sustainable development goals (SDGs) and their importance.**
7. **Explain global governance and decision-making processes.**

Section O: Area Studies (7 Questions)

1. **Define area studies and its significance in social sciences.**
2. **Explain the study of regional politics and economic systems.**
3. **Discuss cultural and historical aspects of a specific region (choose one).**
4. **Explain the impact of geography on regional development.**
5. **Describe the role of regional organizations like ASEAN, EU, or SAARC.**
6. **Discuss migration patterns and demographic trends in a region.**
7. **Analyze a case study of regional conflict and resolution.**

⍰ Ultimate Social Sciences Test (College Level)

Instructions:

- **Attempt all questions.**
- **Answer theory questions with examples and explanations.**
- **Use maps, diagrams, or tables where needed.**
- **Covers History, Geography, Political Science, Civics, Sociology, Anthropology, Psychology, Social Work, Population & Gender Studies, Cultural Studies, Peace & Conflict Studies, International & Global Studies, Area Studies.**

Section A: History (World, Regional, Islamic) (7 Questions)

1. **Describe the main causes and effects of World War I.**
2. **Explain the rise and fall of the Roman Empire.**
3. **Discuss the major achievements of the Islamic Golden Age.**
4. **Explain the impact of colonialism on South Asia.**
5. **Describe the French Revolution and its significance in world history.**

6. **Discuss the Mughal Empire's administrative and cultural contributions.**
7. **Explain the causes and effects of the Cold War.**

Section B: Geography (Physical, Human, Economic) (7 Questions)

1. **Explain the differences between physical, human, and economic geography.**
2. **Describe the process of plate tectonics and its impact on earth's surface.**
3. **Explain factors affecting population distribution.**
4. **Discuss urbanization and its socio-economic effects.**
5. **Explain the concept of sustainable development in geography.**
6. **Describe major climatic zones of the world.**
7. **Explain the role of natural resources in economic development.**

Section C: Political Science (7 Questions)

1. **Define political science and its importance.**
2. **Explain the differences between democracy, monarchy, and authoritarian regimes.**
3. **Discuss the role of political parties in modern democracies.**
4. **Explain the concept of separation of powers.**
5. **Discuss the significance of civil rights and liberties.**
6. **Explain the concept of governance and good governance indicators.**
7. **Analyze the role of international organizations like the UN in global politics.**

Section D: Civics (7 Questions)

1. **Explain the fundamental rights and duties of citizens.**
2. **Describe the structure of a federal government.**
3. **Explain the electoral process in a democratic system.**
4. **Discuss the role of local governments in public administration.**
5. **Explain the importance of rule of law in society.**
6. **Discuss civic responsibilities in promoting social harmony.**
7. **Explain the role of civil society organizations in public life.**

Section E: Sociology (7 Questions)

1. **Define sociology and its major areas of study.**
2. **Explain the concepts of culture, norms, and values.**
3. **Discuss the role of social institutions in society.**
4. **Explain the impact of social stratification on communities.**
5. **Describe the difference between primary and secondary social groups.**
6. **Discuss the concept of social change and its drivers.**
7. **Explain the role of sociology in understanding modern social issues.**

Section F: Anthropology (7 Questions)

1. **Define anthropology and its main subfields.**
2. **Explain the importance of cultural anthropology in understanding societies.**
3. **Discuss the study of human evolution and its methods.**
4. **Explain the concept of ethnography and its application.**
5. **Describe the role of archaeology in studying past civilizations.**
6. **Discuss the significance of kinship and social organization in anthropology.**
7. **Explain applied anthropology and its relevance in modern society.**

Section G: Psychology (7 Questions)

1. **Define psychology and its main branches.**
2. **Explain the difference between clinical and developmental psychology.**
3. **Discuss major psychological theories: Freud, Piaget, and Skinner.**
4. **Explain cognitive processes: perception, memory, and learning.**
5. **Describe the role of emotions and motivation in human behavior.**
6. **Explain psychological disorders and their classification.**
7. **Discuss the importance of psychological research in society.**

Section H: Social Work (7 Questions)

1. **Define social work and its objectives.**
2. **Explain the role of social workers in communities.**
3. **Discuss methods of social work: casework, group work, and community work.**
4. **Explain the importance of ethics in social work practice.**

5. **Describe challenges faced by social workers in modern society.**
6. **Discuss social welfare policies and their impact.**
7. **Explain the role of NGOs in social development.**

Section I: Population Studies (7 Questions)

1. **Define population studies and its significance.**
2. **Explain factors affecting population growth.**
3. **Discuss the demographic transition model.**
4. **Explain the social and economic impacts of overpopulation.**
5. **Discuss population policies in developing countries.**
6. **Explain migration and its effects on society.**
7. **Describe methods for population data collection.**

Section J: Gender Studies (7 Questions)

1. **Define gender studies and its importance in social sciences.**
2. **Explain the difference between sex and gender.**
3. **Discuss gender roles and stereotypes in society.**
4. **Explain the impact of gender inequality on development.**
5. **Describe women's empowerment initiatives globally.**
6. **Discuss the role of education in promoting gender equality.**
7. **Explain how gender studies intersect with other social sciences.**

Section K: Cultural Studies (7 Questions)

1. **Define culture and its components.**
2. **Explain the role of language, religion, and traditions in culture.**
3. **Discuss cultural diversity and its importance.**
4. **Explain cultural diffusion and globalization effects.**
5. **Describe the impact of mass media on cultural identity.**
6. **Discuss the role of festivals and rituals in society.**
7. **Explain cultural heritage preservation methods.**

Section L: Peace & Conflict Studies (7 Questions)

1. **Define peace studies and conflict studies.**
2. **Explain the types of conflicts: interpersonal, communal, international.**
3. **Discuss conflict resolution strategies and negotiation techniques.**
4. **Explain the role of diplomacy in maintaining peace.**
5. **Discuss the impact of war on societies.**
6. **Describe peace-building initiatives by international organizations.**
7. **Explain the importance of human rights in conflict prevention.**

Section M: International Relations (7 Questions)

1. **Define international relations and its scope.**
2. **Explain theories of international relations: realism, liberalism, constructivism.**
3. **Discuss the role of the United Nations in global governance.**
4. **Explain diplomacy and foreign policy formulation.**
5. **Discuss international trade and economic cooperation.**
6. **Explain the role of treaties and international law.**
7. **Analyze a recent international conflict and its resolution efforts.**

Section N: Global Studies (7 Questions)

1. **Define global studies and its interdisciplinary approach.**
2. **Explain globalization and its effects on economies and cultures.**
3. **Discuss global challenges: climate change, poverty, and migration.**
4. **Explain the role of international organizations in addressing global issues.**
5. **Discuss the impact of global media on society.**
6. **Describe sustainable development goals (SDGs) and their importance.**
7. **Explain global governance and decision-making processes.**

Section O: Area Studies (7 Questions)

1. **Define area studies and its significance in social sciences.**
2. **Explain the study of regional politics and economic systems.**
3. **Discuss cultural and historical aspects of a specific region (choose one).**
4. **Explain the impact of geography on regional development.**

5. **Describe the role of regional organizations like ASEAN, EU, or SAARC.**
6. **Discuss migration patterns and demographic trends in a region.**
7. **Analyze a case study of regional conflict and resolution.**

⍰ Ultimate Combined Test: Economics, Religious Studies, Humanities, Arts, Media, Agriculture, Law, Physical Education, Vocational Skills, Modern Subjects, Special Education

Instructions:

- **Attempt all questions.**
- **Provide explanations, examples, and diagrams where applicable.**
- **For applied topics (Business, Agriculture, Law, Media, Vocational Skills), give practical examples or case studies.**
- **Covers all subjects from Economics & Commerce to Special & Support Subjects.**

Section 1: Economics, Commerce & Business (5 Questions per topic)

Economics (Micro, Macro, Development)

1. **Explain the difference between microeconomics and macroeconomics with examples.**
2. **Discuss the law of demand and supply and illustrate with a graph.**
3. **Explain the concept of opportunity cost in economic decision-making.**
4. **Discuss the role of government in economic development.**
5. **Explain inflation, its causes, and effects on the economy.**

Business Studies

1. **Define business and explain its main objectives.**
2. **Discuss different types of business organizations (sole proprietorship, partnership, corporation).**
3. **Explain the importance of business ethics.**
4. **Describe the role of entrepreneurship in economic growth.**

5. **Discuss the impact of technology on modern businesses.**

Commerce

1. **Define commerce and its components.**
2. **Explain the difference between trade and commerce.**
3. **Discuss e-commerce and its advantages.**
4. **Explain the role of banking in commerce.**
5. **Describe international trade and its benefits.**

Accounting & Financial Accounting

1. **Explain the difference between financial and cost accounting.**
2. **Describe the accounting cycle with examples.**
3. **Explain the concept of double-entry bookkeeping.**
4. **Prepare a basic trial balance from given transactions.**
5. **Explain depreciation methods and their applications.**

Cost Accounting & Management Accounting

1. **Define cost accounting and its objectives.**
2. **Explain fixed, variable, and semi-variable costs.**
3. **Discuss budgeting and its importance in management.**
4. **Explain break-even analysis with a numerical example.**
5. **Describe performance evaluation using standard costing.**

Business Management & HRM

1. **Define management and explain its functions.**
2. **Discuss the principles of scientific management by Taylor.**
3. **Explain the importance of human resource planning.**
4. **Describe methods of employee recruitment and selection.**
5. **Discuss performance appraisal techniques.**

Marketing & Entrepreneurship

1. **Define marketing and explain the marketing mix (4Ps).**
2. **Discuss market segmentation and targeting strategies.**
3. **Explain the role of branding in business success.**
4. **Define entrepreneurship and its characteristics.**
5. **Discuss challenges faced by entrepreneurs in emerging markets.**

E-Commerce & Banking Studies

1. **Explain the difference between traditional commerce and e-commerce.**
2. **Discuss the role of digital payment systems in e-commerce.**
3. **Explain internet banking and mobile banking.**
4. **Discuss cybersecurity challenges in online transactions.**
5. **Describe the role of banks in economic development.**

Finance, Investment, Taxation, Auditing

1. **Define finance and its types (personal, corporate, public).**
2. **Discuss types of investment options and their risks.**
3. **Explain progressive and regressive taxation.**
4. **Discuss the objectives of auditing.**
5. **Explain corporate financial planning and risk management.**

Business Law & Corporate Governance

1. **Define business law and its importance.**
2. **Explain the key provisions of contract law.**
3. **Discuss company law basics and corporate governance principles.**
4. **Explain legal compliance in business operations.**
5. **Describe intellectual property rights and their significance.**

Supply Chain Management & Logistics

1. **Define supply chain management and its components.**
2. **Discuss the importance of inventory management.**
3. **Explain the role of logistics in global trade.**
4. **Discuss transportation modes and their efficiency.**
5. **Describe modern supply chain technologies (like RFID, IoT).**

Section 2: Religious & Ethical Studies (5 Questions per topic)

Islamic Studies, Islamiyat, Quran Studies

1. **Explain the importance of Shahadah in Islam.**
2. **Discuss the core teachings of the Quran.**
3. **Explain the significance of Salah (prayer) in daily life.**
4. **Describe the life of Prophet Muhammad .snotiubirtnoc sih dna ﷺ**
5. **Explain the role of Zakat in social welfare.**

Hadith Studies & Fiqh

1. **Define Hadith and explain its categories.**
2. **Discuss the science of Hadith authentication (Ilm al-Hadith).**
3. **Explain the sources and principles of Fiqh.**
4. **Describe the differences between Sunni and Shia Fiqh.**
5. **Explain practical application of Fiqh in daily life.**

Islamic History & Civilization

1. **Discuss the major achievements of early Islamic caliphates.**
2. **Explain the spread of Islam in South Asia.**
3. **Describe the contributions of Islamic scholars to science.**
4. **Discuss cultural developments during the Abbasid era.**
5. **Explain the role of Islamic civilization in art and architecture.**

Islamic Economics, Law & Ethics

1. **Define Islamic economics and its principles.**
2. **Explain prohibition of riba (interest) in Islamic finance.**
3. **Discuss the role of Shariah in business ethics.**
4. **Explain Islamic inheritance laws (Faraid) with examples.**
5. **Discuss ethical values promoted in Islam.**

Seerah (Life of Prophet ﷺ) & Other Religious/Ethics

1. **Describe the early life of Prophet Muhammad ﷺ.**
2. **Explain the significance of Hijrah (migration).**
3. **Discuss key battles in the Prophet's life and their outcomes.**
4. **Explain comparative religion and interfaith understanding.**
5. **Discuss the role of ethics and moral philosophy in society.**

World Religions, Peace Studies & Values Education

1. **Compare the teachings of Buddhism, Christianity, and Islam.**
2. **Discuss the role of religion in promoting peace.**
3. **Explain ethical principles common across major religions.**
4. **Describe value-based education and its importance.**
5. **Explain conflict resolution through ethical practices.**

Section 3: Humanities & Philosophy (5 Questions per topic)

Philosophy, Logic, Ethics

1. **Define philosophy and its main branches.**
2. **Explain deductive vs inductive reasoning.**
3. **Discuss ethical theories: utilitarianism, deontology, virtue ethics.**
4. **Explain the concept of moral relativism.**
5. **Discuss the role of philosophy in modern society.**

Aesthetics & Critical Thinking

1. **Define aesthetics and its relevance to arts.**
2. **Explain elements of critical thinking.**
3. **Discuss logical fallacies and examples.**
4. **Explain the role of reasoning in decision-making.**
5. **Discuss the connection between ethics and aesthetics.**

Research Methodology & Academic Writing

1. **Explain qualitative and quantitative research methods.**
2. **Discuss hypothesis formulation and testing.**
3. **Describe data collection techniques (surveys, interviews, observation).**
4. **Explain citation styles and plagiarism prevention.**
5. **Discuss structure of an academic research paper.**

Communication Studies & Media Ethics

1. **Define communication and its types.**
2. **Explain barriers to effective communication.**
3. **Discuss the importance of media ethics.**
4. **Describe interpersonal and mass communication.**
5. **Explain role of communication in organizations.**

Section 4: Arts, Design & Creative Studies (5 Questions per topic)

Fine Arts, Drawing, Painting, Sculpture

1. **Explain the principles of visual arts.**
2. **Discuss elements of design in drawing.**
3. **Describe techniques in painting (watercolor, oil, acrylic).**
4. **Explain sculpture materials and methods.**

5. **Discuss art as a form of cultural expression.**

Graphic, Fashion, Textile & Interior Design

1. **Explain basic principles of graphic design.**
2. **Discuss current trends in fashion design.**
3. **Describe textile printing and weaving techniques.**
4. **Explain interior design planning and aesthetics.**
5. **Discuss the role of design in functional and aesthetic contexts.**

Industrial Design, Animation & Film Studies

1. **Define industrial design and its impact on product development.**
2. **Explain animation techniques (2D vs 3D).**
3. **Discuss principles of cinematography.**
4. **Describe storyboard creation for films or animations.**
5. **Explain the influence of media on society through film.**

Photography, Music & Drama

1. **Explain elements of photography (composition, lighting, angles).**
2. **Discuss music theory basics for vocal and instrumental music.**
3. **Describe drama and theatre elements (script, stage, performance).**
4. **Explain performing arts as a tool for cultural education.**
5. **Discuss creative writing and literature analysis.**

Section 5: Media, Communication & Journalism (5 Questions per topic)

1. **Define journalism and its types.**
2. **Explain the role of mass media in society.**
3. **Discuss advertising strategies and ethics.**
4. **Describe public relations and communication campaigns.**
5. **Explain digital media trends and content creation practices.**

Section 6: Agriculture & Environment (5 Questions per topic)

1. **Explain crop science and crop rotation benefits.**

2. **Describe horticultural practices for fruits and vegetables.**
3. **Explain livestock management basics.**
4. **Discuss environmental management and sustainable agriculture.**
5. **Describe climate change impacts on agriculture.**

Section 7: Law, Governance & Public Service (5 Questions per topic)

1. **Explain constitutional principles and citizen rights.**
2. **Discuss international law and treaties.**
3. **Describe human rights protection mechanisms.**
4. **Explain criminology and forensic studies basics.**
5. **Discuss public administration and governance practices.**

Section 8: Physical, Sports & Health Education (5 Questions per topic)

1. **Define physical education and its benefits.**
2. **Explain sports science and exercise physiology.**
3. **Discuss coaching principles and athlete development.**
4. **Explain yoga and martial arts benefits.**
5. **Describe health education and lifestyle management.**

Section 9: Vocational & Technical Skills (5 Questions per topic)

1. **Explain principles of electrical, mechanical, and automotive technology.**
2. **Discuss plumbing, welding, and HVAC systems.**
3. **Explain renewable energy technologies: solar, wind.**
4. **Describe carpentry and construction techniques.**
5. **Discuss modern tools and safety practices in technical trades.**

Section 10: Modern & Emerging Subjects (5 Questions per topic)

1. **Explain artificial intelligence ethics and applications.**
2. **Discuss robotics and automation in industries.**
3. **Describe nanotechnology and biotechnology basics.**
4. **Explain quantum computing (intro) and potential applications.**
5. **Discuss smart cities, digital transformation, and innovation studies.**

Section 11: Special & Support Subjects (5 Questions per topic)

1. **Define special and inclusive education.**
2. **Explain learning psychology and guidance counseling.**
3. **Discuss career planning and life skills development.**
4. **Explain leadership studies and personality development.**
5. **Discuss emotional intelligence and its importance in personal growth.**